Assessment of K-Means Algorithm to Evaluate Nearshore Wave Climate

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***Abstract*—Accidents near ports have increased due to the ongoing expansion of maritime trade. These accidents have various causes, including adverse weather conditions. Accurate wave climate forecasts can help mitigate the risks associated with marine accidents. While numerical models are commonly used for obtaining nearshore wave climate forecasts, their high computational cost makes them impractical for wave climate forecasting and nowcasting. Artificial neural networks (ANNs) offer a potential solution to this limitation. However, existing ANNs have primarily focused on specific single points within the study areas, such as piers and port entrances. Enhancing early-warning strategies requires a broader understanding of the wave climate across larger areas. Thorough examinations of extensive areas with varying physical attributes can result in significant computational time requirements. The main objective of this study is to evaluate a clustering technique able to identify homogeneous areas to improve future applications of ANNs to assess nearshore wave characteristics in actual situations. The area around the port of Augusta (Sicily), one of the most important ports in Italy, serves as a case study in this article. Results show an optimal performance by applying the clustering algorithm K-means, capable of capturing the wave climate characteristics of the study area.**

***Index Terms*—Clustering algorithms, K-means, maritime accidents, SWAN, wave climate.**

I. Introduction

MARITIME transportation is essential for global trade and commerce, as it is responsible for carrying over 80% of the world's goods by volume. Over the past ten years, it has grown significantly at a rate of about 3% each year [1]. However, like any mode of transportation, maritime transportation is not without risks, and accidents can occur, sometimes with devastating consequences. Accidents in maritime transportation can result from a variety of causes, including human error, equipment failure, weather conditions, and natural disasters. One of the most common causes of maritime accidents is human error, such as navigational blunders, miscommunication between crew members, or the mishandling of the vessel [2]. Equipment failure can lead to accidents, such as mechanical failure in the engine or a malfunctioning in the steering system. Weather conditions can also be a significant factor in maritime accidents, especially in rough seas or storms. These conditions can make it challenging to navigate and control the vessel, leading to accidents, such as collisions, grounding, or capsizing [3]. Natural disasters, such as hurricanes, typhoons, or tsunamis, can also cause severe damage to vessels. To prevent and mitigate maritime accidents, the scientific community focuses on the risks of maritime operation in order to identify the best mitigation strategies by adopting well-established approaches and new promising tools [4]. However, accidents can still occur despite these measures, highlighting the need for continued improvement in safety practices and technologies.

In this context, knowing the local weather and sea conditions well in advance is undoubtedly helpful for reducing the risk of accidents, thus aiding ship routing decisions.

However, in deep water, where there is a small or negligible interaction between waves and the sea bottom, sea state forecasting is usually available. The knowledge of nearshore wave climate is crucial for the determination of operational conditions in port areas. As a result, sea-state data should be propagated towards the coast to obtain nearshore information, taking into account the wave transformation processes. Numerical models, either based on spectral approaches or on the resolution of Boussinesq equation, are frequently used to obtain sea state forecasts toward the coast. These models, however, have expensive processing requirements and cannot be used within nowcasting and forecasting system [5]. Artificial neural networks (ANNs) can be used to overcome this restriction and effectively retrieve the necessary metocean data. In several studies, ANNs have been applied to forecast sea states [6], [7], [8] and storm surge [9], [10], [11]. However, existing ANNs have been developed to focus on specific points of interest, such as piers and port entrances, which means that ANNs are designed to retrieve metocean data only at these specific points and and may not take into consideration other areas where the data could be relevant for risk mitigation. In order to improve the efficiency and accuracy of early-warning

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systems, there is a need for more extensive knowledge on the wave climate across a wider area. This broader data could help to identify potential risks and mitigate damages. Recent studies have shown that artificial intelligence can accurately predict nearshore wave patterns throughout entire regions [12], [13]. However, these studies only considered areas with uniform bars or circular shoals, which do not reflect morphological conditions.

In this context, the primary aim of this research is to address the critical challenge of accurately assessing nearshore wave characteristics under real morphological circumstances, particularly focusing on the area surrounding the port of Augusta in Sicily, Italy. This objective is paramount due to the significant impact that local weather and sea conditions can have on maritime transportation safety in this region. By understanding and predicting nearshore wave patterns more effectively, the goal is to enhance early warning systems and risk mitigation strategies for maritime operations. To achieve this aim, a novel approach that involves two main steps is employed [14]. First, numerical models are utilized to reconstruct nearshore wave climate characteristics, taking into account the diverse morphological conditions present in the Augusta area. However, the computational requirements of analyzing extensive areas with varied physical attributes can be substantial. Therefore, the second step involves implementing clustering algorithms to identify areas with homogeneous conditions, thereby reducing computational demands and improving the efficiency of subsequent applications of ANNs for each cluster. Indeed, applying clustering algorithms within the methodology serves several crucial purposes. Clustering algorithms like K-means can group similar conditions based on geographical and physical attributes. This grouping allows for a more focused analysis and modeling approach tailored to each cluster's specific characteristics. Moreover, identifying areas with similar conditions reduces the computational burden by enabling more targeted modelling efforts within each homogeneous cluster rather than applying the model uniformly across the entire area. Each cluster represents a distinct set of environmental conditions, allowing ANNs to learn and predict nearshore wave characteristics more accurately within those specific contexts. This approach enhances the predictive capabilities of ANNs by focusing on relevant data subsets rather than attempting to model the entire heterogeneous area uniformly. Consequentially, clustering helps identify critical areas with similar wave characteristics, enabling more precise predictions supporting timely warnings and proactive risk management measures. This capability is essential for improving maritime transportation safety and mitigating potential hazards from local weather and sea conditions. By focusing on the specific case study of the port of Augusta, which encompasses diverse morphological conditions leading to various sea state phenomena, the research aims to provide valuable insights and tools that can be applied to similar port regions globally. Ultimately, the overarching goal is to contribute to the advancement of safety practices and technologies in maritime transportation through more accurate and comprehensive assessments of nearshore wave characteristics. Preliminary results were presented in Castro et al. [15], but extended ones are shown in this article, in which results of the whole area of Augusta are considered.

The rest of this article is organized as follows. Section II introduces the clustering algorithms and the methodology used in the present work to implement the K-means algorithm; a description of the area of study is briefly described in Section III; the results of the proposed methodology are provided in Section IV. Finally, Section V concludes this article.

II. Clustering Algorithms

Nowadays, a large amount of different data are available at a number of locations and from a variety of sources. However, the majority of data are gathered in arbitrary formats and classifications, making it challenging to analyze them, particularly when the properties of the data items are unknown [16]. In this sense, data mining's main objective is to extract useful information from acquired data, in which the cluster analysis is able to manage the component of data mining that deals with the proper structuring of unlabeled data. Data clustering is the process of grouping such data in a meaningful way. The objective is to cluster unlabeled data so that data items with comparable traits and attributes are grouped in one cluster and that the similarities of data objects within a cluster are greater than those of data objects in other clusters. In other words, to ensure higher intracluster and lower intercluster similarity, unlabeled data are categorized by data clustering analysis [17]. Accordingly, clustering analysis differs from discriminant analysis; the first one is an unsupervised classification, whose aim that it does not require labeled data or prior knowledge about the data, whereas the second one is a supervised classification, which the task is to categorize a newly discovered, unlabeled pattern from a set of labeled and preclassified patterns [18].

Historically, there are two main categories of clustering techniques: hierarchical and partitioning clustering. The first one decomposes the dataset into a hierarchy of groups and this decomposition is represented by a dendrogram, which is a binary union tree where the leaves represent each data of the dataset and the root represents the entire dataset [19]. The height of the dendrogram typically reflects the distance between each pair of items or clusters, or an object and a cluster, and the intermediate nodes describe the proximity of the objects. [20]. There are two types of hierarchical approaches: agglomerative (or bottom–up) and divisive (or top–down) [21]. In the first approach, each object is considered as a separate cluster. The proximity between two data is then determined by choosing a distance metric, and the closest pairs are combined into a single cluster. Up until all the data points are integrated to form a single cluster, this procedure continues iteratively. As a result, the dendogram produced by the agglomerative clustering algorithm will have leaf nodes that are single article clusters and a root node that contains all of the articles [22]. In contrast to agglomerative methods, in the divisive approach all data start in one cluster: that is, a root node. The method then attempts to partition the datasets until they reach a leaf node carrying a single object [23]. Hierarchical clustering algorithms have several advantages, such as a high applicability to any attribute type and also they can represent

nested clusters. However, the majority of hierarchical algorithms do not go back and refine previously created clusters in order to improve the method and have high computation costs [24]. On the other hand, partitioning clustering creates a single partition of the initial dataset, rather than creating a hierarchy structure. In particular, it divides x data into k clusters according to an evaluation criteria, with $k \leq x$. A centroid is used to represent each cluster and each data belongs to only one cluster [25].

One of the most widely used partitioning clustering algorithm, thanks to its easy implementation, is the *K-means* one, which is a centre-based algorithm. It was created forty years ago by MacQueen [26] and seeks to identify k clusters that minimize the sum of the square error for each cluster. A cluster is represented by its centroid, which is the average of all the points that make up the cluster. Varying the location of these centroids also produces a variation in the results themselves and consequently the best setup is to place them at a distance from each other. Then, the algorithm associates the data to the nearest centroid. Specifically, K-means starts with a random initial partition and recomputes the centroids, which change their position and reassigns the data to clusters until a convergence criterion is met.

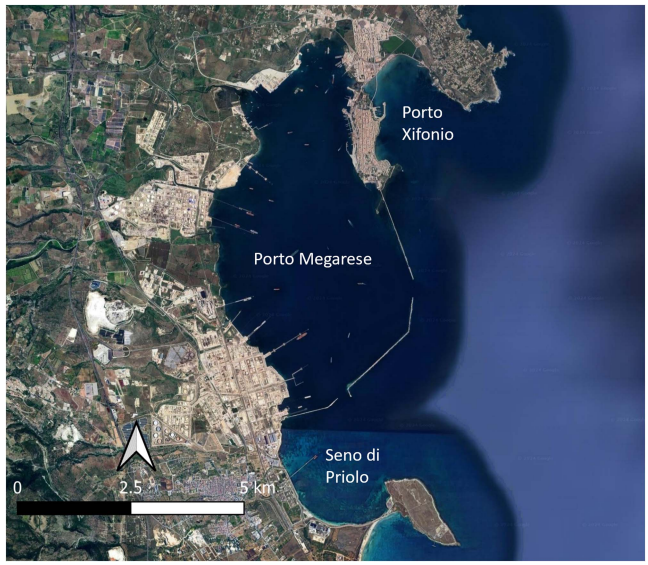


Fig. 1. Port of Augusta.

A. Implementation of the K-Means Algorithm

The K-means algorithm was applied in the present work, by varying the input parameters. Indeed, in the implementation of the algorithm, there are two main input parameters that need to be specified: the *distance metric*, which is the measure used to calculate the similarity or dissimilarity between data and k, which represents the number of clusters that the algorithm should generate. In particular, users need to select an appropriate distance metric that aligns with the characteristics of the data being clustered and need to determine the desired number of clusters, based on their understanding of the data or the problem domain. Then, the determination of these parameters might be challenging in a dataset with high density and complexity.

In this study, various distance metrics were compared, including Euclidean, City-block, Cosine, and Correlation distances. Euclidean distance, commonly used in K-means clustering [27], allows the inclusion of new data without impacting the analysis. However, it is not scale-invariant, meaning that attributes with larger ranges can disproportionately influence the distance calculation [28]. City-block distance offers shorter computation times than Euclidean distance but is sensitive to outliers [29]. Cosine distance, which ignores vector magnitudes, may result in the loss of important information, making it unsuitable for datasets where vector magnitude is significant [30]. Correlation distance, on the other hand, effectively reveals shape or pattern dissimilarities between data points rather than quantifying magnitude differences [20]. One of the most challenging problems in cluster analysis is determining the optimal number of clusters, as it depends on various factors, such as the nature of the data. The goal is to find a balance where increasing the number of clusters beyond the optimal point does not yield significant additional insights while keeping the number of clusters low enough to maintain interpretability. Criteria, such as Calinski–Harabasz [31], Davies–Bouldin [32], Silhouette [33], and Total Variance [34], were compared to address this issue. These criteria involve calculating an equation for various numbers of clusters and selecting the value that minimizes or maximizes a specific criterion. For instance, a higher Calinski–Harabasz criterion ratio indicates better data partitioning, with the maximum value representing the optimal number of clusters. The Davies–Bouldin criterion evaluates clustering quality by comparing each cluster to every other cluster based on a similarity function, considering the worst-case scenarios for each cluster. It divides the measure of compactness by the measure of separation, meaning that as clusters become more compact and well-separated, the criterion value decreases [35]. A high Silhouette value suggests that a point is well-matched to its cluster and poorly matched to other clusters. The clustering solution is considered appropriate if most points have high Silhouette values. However, if many points have low or negative values, the clustering solution may have too many clusters [33]. Lastly, the Total Variance criterion indicates that total variance decreases as clusters increase. Therefore, identifying where the total variance curve flattens out can determine the optimal number of clusters.

III. Description of the Port of Augusta

The port of Augusta is partially natural and is located on the Eastern coast of Sicily and is bordered by a rocky headland to the North and the Magnisi peninsula to the South. The Augusta bay is made up of the following three large sections (see Fig. 1).

1) *Porto Xifonio*, which is the part of the sea between Punta Izzo and Punta Carcarella, in which the refraction phenomena are relevant.
2) *Porto Megarese*, that is the area sheltered by the breakwaters, which is dominated by diffracted waves and local generated waves.
3) *Seno di Priolo*, which is the area from the southern Megarese breakwwaters and Magnisi Peninsula.

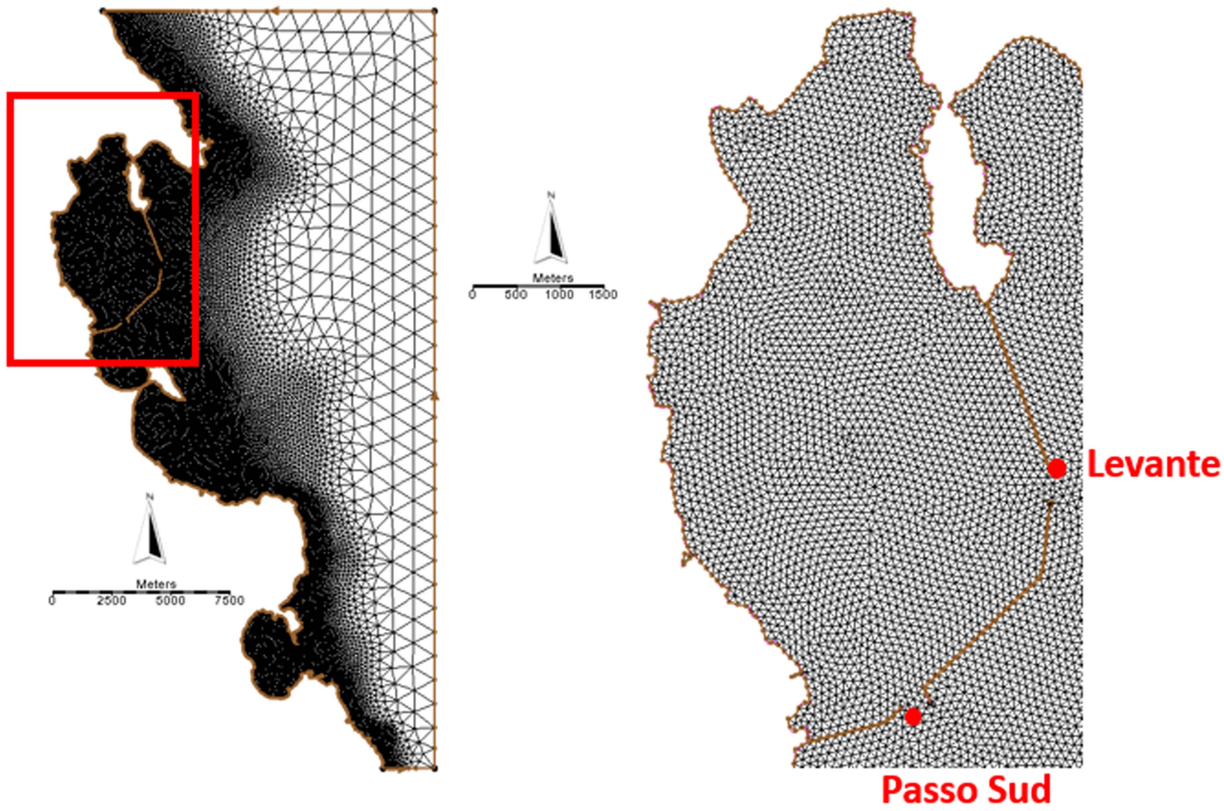


Fig. 2. Computational grid used to simulate the wave propagation from offshore to the study area: (a) complete grid; (b) detail of the grid within the harbour.

The harbor covers an area of approximately 23 km^2 and is protected by 6.5 km of breakwaters, with two entrances: Levante and Passo Sud, which are exposed to waves from the first and second quadrants. The Levante entrance is around 480 m wide, while Passo Sud is approximately 260 m wide. The average depth of the sheltered area is about 15 m, with the maximum depth being 21.8 m in the central area of the sheltered area. The operative piers have an average depth of around 10 m. The harbor's large size allows for a maximum fetch length of approximately 4.5 km along the east–west direction and 8 km along the north–south direction. However, the harbor's vast basin, combined with local winds, can cause significant wave generation inside the harbor, posing threats to maneuvering and moored ships. Furthermore, the harbor has a high volume of maritime traffic, with 40% of accidents resulting from collisions with buoys or docks, 25% from ship-to-ship collisions, and 30% from sinking and grounding [36]. Adverse weather conditions, due to high wind speeds, and the muddy seabed of the sheltered area frequently cause the displacement of anchored or moored ships, leading to collisions with other vessels.

IV. Result and Discussion

With the aim of contributing to the development of a novel method for evaluating real-time nearshore waves conditions, the proposed methodology's initial phase consists of analysing wind and wave characteristics at the study area to create a database of potential scenarios. These scenarios are then used as input for advanced numerical modeling technique. Second, through an expert approach, the entire area is divided into macro-areas according to the output data of the numerical model. Lastly, a clustering algorithm is implemented for each subareas in order to improve the evaluation of nearshore wave conditions.

A. Reconstruction of Wave Climate Characteristics in the Port Area

The dataset used as input for the clustering algorithm was developed by using the wave propagation model SWAN (*Simulating Wave Nearshore*), developed by the Delft University of Technology [37]. This model is a third-generation wave model that enables the prediction of wave characteristics in coastal areas, lakes, and estuaries that are generated by wind and the simulation of various wave propagation phenomena, such as diffraction, refraction, shoaling, and reflection. Two calculation grids are necessary for the model: a wave spectrum grid and a geographical grid. The wave spectrum grid is used to discretise the spectrum of each sea state that propagates toward the coast, while the geographical grid is defined based on the bathymetry and should have higher resolution in areas with rapidly changing wave climates. The unstructured calculation grid used in this study consists of 30 080 triangular cells connected by 15 607 nodes, allowing for greater grid density in complex areas, such as those near the coast (see Fig. 2). In addition, unstructured

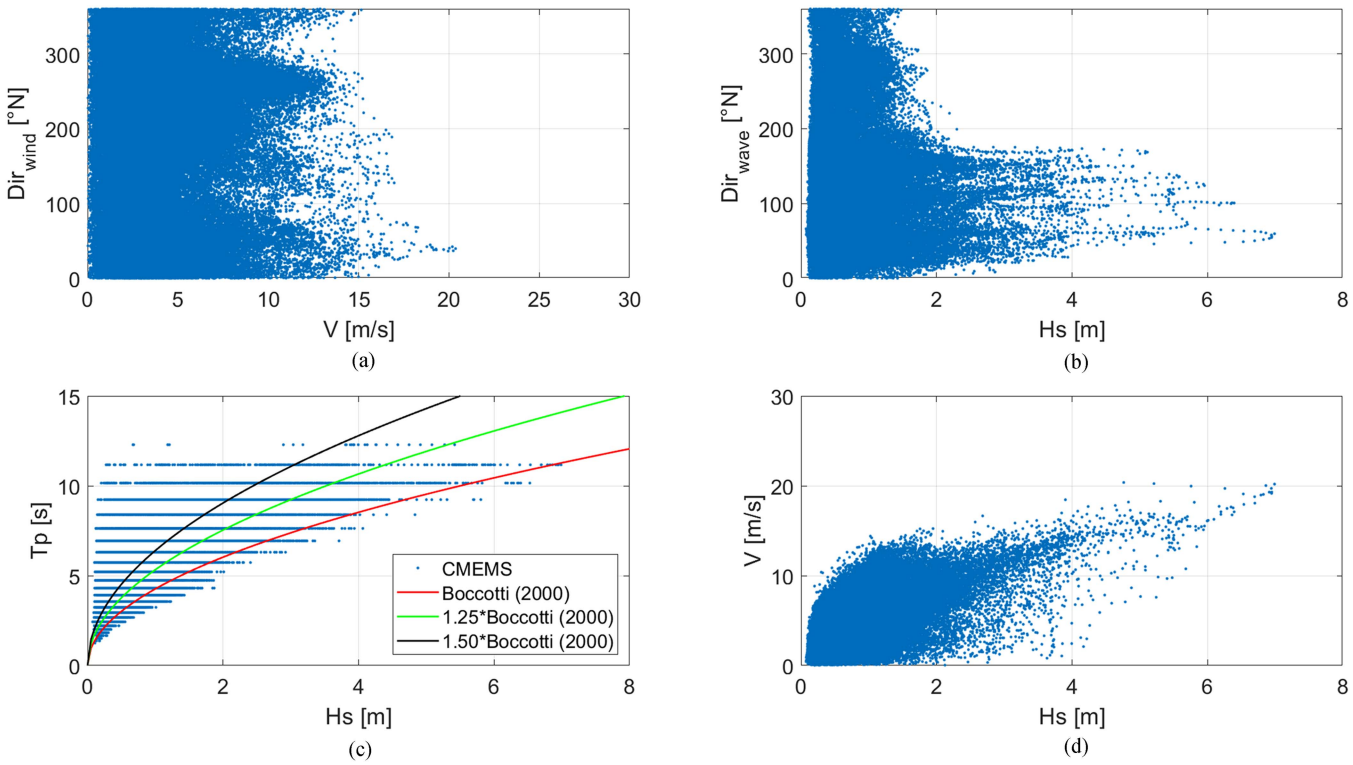


Fig. 3. Analysis of the wind and wave data provided by the ECMWF and CMEMS from 2006 to 2019, respectively: (a) Wind direction as a function of the wind velocity; (b) wave direction as a function of the Hs; (c) comparison between the peak wave period estimated with Boccotti [39] and that provided by the CMEMS; (d) wind velocity as a function of the Hs.

grids offers a simplified representation of the investigating area with relatively high precision while requiring fewer grid points than regular grids [38].

To determine potential scenarios for the chosen location, an evaluation of wind and wave climate was carried out. In the absence of direct measurements, hourly data from 2006 to 2019 from the European Centre for Medium-Range Weather Forecasts ERA5 (ECMWF - ERA5) and the Copernicus Marine Environment Monitoring Service (CMEMS) were utilized. Wind data was obtained from the ECMWF, while wave data was obtained from the CMEMS. The analysis of these datasets aided in defining likely offshore situations. Fig. 3 displays a summary of the two datasets in a synthetic manner. It is apparent from the figure that waves arriving from the open sea, with directions ranging from 0°N to 180°N, resulted in greater significant wave heights (Hs) and uncorrelated with local wind direction.

Considering the above and according to Iuppa et al. [40], SWAN was used to simulate 146 875 scenarios, by varying the wind properties (i.e., velocity V and direction $\mathrm{Dir}_{\mathrm{wind}}$) as well as the wave characteristics (i.e., Hs, mean direction $\mathrm{Dir}_{\mathrm{wave}}$, and peak wave period Tp). The peak wave period was calculated according to the formula proposed by Boccotti [39]

$$T_p = 8.5\pi\sqrt{\frac{Hs}{4g}} \tag{1}$$

where g is the gravity acceleration [m/s^2]. However, this formula is strictly valid just for the peak of a wave storm. In order to take into account various wave conditions, two additional values were considered by increasing the value of Tp obtained using such a relationship by 25% and 50%, respectively [see Fig. 3(c)].

The analyzed area is sheltered by waves coming from directions in the range 155–15°N. Therefore, two cases have been simulated. The first one refers to sea states with the wind direction coming from 0°N to 360°N and the wave directions coming from 15–155°N. Case 2 refers to sea states generated by local winds coming from 180°N to 360°N. For each SWAN simulation, the execution time was about 10 minutes using a computer with a RAM of 32GB and a Intel Xeon Silver 4214 CPU @2.20GHz.

Table I summarizes the input data used for the model SWAN.

B. Division of the Study Area Into Macro-Areas

The entire area of Augusta was first divided into eight macro-areas through an expert approach, in order to identify physically homogeneous macro-areas from the viewpoint of nearshore significant wave heights Hs. Indeed, the port breakwaters and the headlands partially shelter some of these areas. Fig. 4 shows the identification of these eight macro-areas: *Costa Pergola*, which is a rocky headland sited to the North of Augusta; *Porto Xifonio*, *Porto Megarese,* and *Seno di Priolo*, which represent the Augusta bay; *Rada di Santa Panagia*, that is located in the

TABLE I
WIND AND WAVE CHARACTERISTICS PROVIDED AS INPUT FOR THE NUMERICAL MODEL SWAN

Case	Variables	Range	Step
Case 1: wind direction coming from 0°–360°N	Dir_{wind}	0°–360°N	10°
	V	5–20 m/s	2.5 m/s
	Hs	0.5–7 m	0.5 m
	Dir_{wave}	15°–155°N	10°
Case 2: wind direction coming from 180°–360°N	Dir_{wind}	180°–360°N	10°
	V	5–15 m/s	2.5 m/s
	Hs	-	-
	Dir_{wave}	-	-

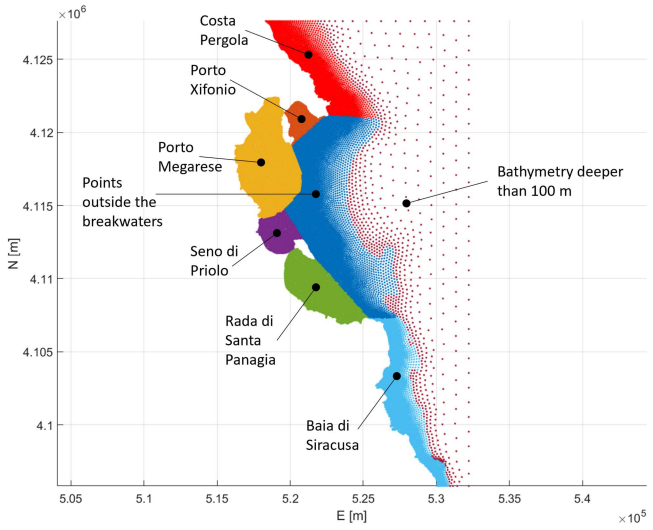


Fig. 4. Identification of eight macro-areas through an expert approach.

North of Syracuse, between Punta Magnisi and Capo Santa Panagia, taking its name from the district of Santa Panagia, which is located nearby; *Baia di Siracusa* that hosts the port of Syracuse; an area, in which the bathymetry is deeper than 100 m and another area, which includes all the points outside the breakwaters of the Augusta bay.

Upon examining the initial clustering, it becomes evident that the study area encompasses multiple areas influenced by common phenomena associated with wave propagation. Indeed, the area of study exhibits notable features, namely, the presence of several bays, such as Porto Xifonio or Seno di Priolo, and sheltered zones that give rise to diffraction phenomena. Moreover, refraction and shoaling phenomena are evident over the entire area.

However, only seven of the presented macro-areas are further investigated as the data points beyond a depth of 100 m are not influenced by wave propagation phenomena like shoaling. Therefore, the macro-area in which the bathymetry is deeper than 100 was not analyzed.

Further clusterization can be obtained through the application of the methodology discussed earlier, i.e., using the K-means algorithm. The input data x for the K-means algorithm consisted of the simulated Hs for each macro-area of the Augusta port, which were obtained using the SWAN model. In the present work

TABLE II
PERFORMANCE INDEXES CALCULATED FOR CITY-BLOCK, EUCLIDEAN, COSINE, AND CORRELATION DISTANCE FOR RADA DI SANTA PANAGIA

	Cityblock	Euclidean	Cosine	Correlation
RMSE [m]	0.337	0.405	0.592	0.659
d	0.953	0.871	0.882	0.903
R	0.977	0.918	0.917	0.914

different evaluations were made by varying the distance metric and the number of clusters in order to find the best parameters configuration for each subarea. Therefore, in order to evaluate the performance of the clustering, the root-mean-square error (RMSE), the coefficient of correlation (R), and the index of agreement (d) [41] were calculated. In more detail, within each cluster, a point with attributes similar to those of the centroid was identified, and the Hs associated with this point (X_i) was compared to the Hs values of the other points within the cluster (Y_i). Mathematically, the indexes were calculated as follows:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(X_i - Y_i)^2}{n}} \tag{2}$$

$$R = \frac{\sum_{i=1}^{n}|X_i - \bar{X}||Y_i - \bar{Y}|}{\sqrt{\sum_{i=1}^{n}|X_i - \bar{X}|^2}\sqrt{\sum_{i=1}^{n}|Y_i - \bar{Y}|^2}} \tag{3}$$

$$d = 1 - \frac{\sum_{i=1}^{n}(X_i - Y_i)^2}{\sum_{i=1}^{n}|X_i - \bar{X}| + |Y_i - \bar{X}|} \tag{4}$$

where n is the number of points in a cluster; $\overline{X}$ and $\overline{Y}$ are the means of X_i and Y_i, respectively.

C. Assessing the Distance Metrics

Initially, three clusters were arbitrarily chosen for each macro-area to evaluate the performance of four distance metrics: City-block, correlation, cosine, and Euclidean. As an example for the sake of brevity, only Rada di Santa Panagia area is illustrated here. Fig. 5 presents the results obtained by the algorithm for the Hs values, considering the analysis of 146 875 scenarios using the SWAN model.

At first sight, when considering City-block and Euclidean distances [see Fig. 5(a) and (d), respectively], the cluster boundaries appear to align with the bathymetry of the area. This alignment contributes to a reliable representation of the Hs. This hypothesis is also confirmed by the mean values of the performance indexes. Indeed, the City-block and Euclidean distances provide the best results in terms of RMSE, R, and d, which are summarized in Table II.

In a high-dimensional space, such as the one of the presented dataset, several phenomena arise that can impact the effectiveness and interpretability of distance metrics. One is the "curse of dimensionality," which is a term used to describe the challenges and limitations that arise when working with high-dimensional data [42].

Aggarwal et al. [43] conducted an extensive empirical study to investigate the behavior of different distance metrics in high-dimensional spaces. They found that the Euclidean distance

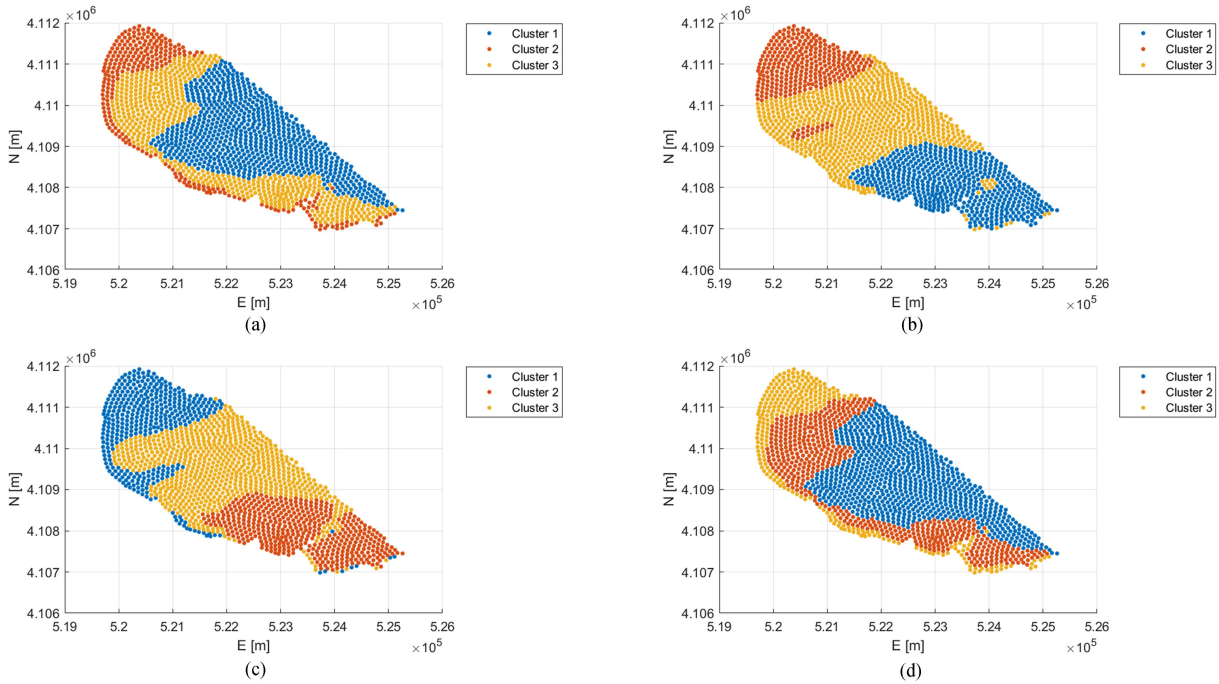


Fig. 5. Application of the clustering algorithm to Rada di Santa Panagia for different distance metrics: (a) City-block, (b) correlation, (c) cosine, and (d) Euclidean.

TABLE III
VALUES OF THE OPTIMAL NUMBER OF CLUSTERS FOR EACH AREA EVALUATED WITH THE CRITERIA: CALISNKI–HARABASZ, DAVIES–BOULDIN, SILHOUETTE, AND TOTAL VARIANCE

	Calinsk–Harabasz	Davies–Bouldin	Silhouette	Total Variance
Costa Pergola	5	2	2	4
Porto Xifonio	2	2	2	8
Porto Megarese	2	2	2	8
Seno di Priolo	4	2	2	8
Rada di Santa Panagia	3	2	2	6
Baia di Siracusa	3	2	2	9

becomes less reliable and may fail to capture meaningful relationships between data points. Moreover, in scenarios where the number of dimensions is 20 or greater, the City-block distance metric tends to show a much clearer distinction between data points compared to the Euclidean distance metric. In other words, the City-block distance metric is better at emphasizing differences between data points in high-dimensional spaces.

Similar results were found in the present study, in which a high-dimensional dataset was used as input data for the clustering technique. The results provided by the algorithm assessment showed that, for each macro-area, the City-block distance metric better represents the distribution of Hs. For this reason, the City-block distance has been chosen as the metric of the algorithm in this article.

D. Evaluation of the Optimal Number of Clusters

To find the optimal number of clusters for each area, the criteria already discussed were applied. In particular, for the Rada di Santa Panagia, the Calinski-Harabasz criterion provides an optimal number of clusters equal to 3; whereas the Davies-Bouldin, and Silhouette criteria indicated an optimal number of clusters of 2 [see Fig. 6(a)–(c)]. In terms of the Total Variance criterion, Fig. 6(d) depicts a decreasing trend in the curve as the number of clusters increases for Rada di Santa Panagia. The graph suggests that the Total Variance metric levels off when the number of clusters reaches around 6, indicating that further increasing the number beyond six may not enhance the clustering performance significantly.

For the other areas, the values of the optimal number of clusters provided by the four criteria are shown in Table III. The outcomes derived from Calinski–Harabasz, Davies–Bouldin, and Silhouette criteria are unsatisfactory due to their lower results. For example, Rada di Santa Panagia area experiences significant spatial variations in wave height due to its large dimensions and nonhomogeneous bathymetry. Therefore, multiple clusters are necessary to identify areas with different wave behavior in terms of Hs. For this reason, the Total Variance criterion provides more plausible results, also according to the performance indexes evaluated for each number of clusters of each area. Specifically,

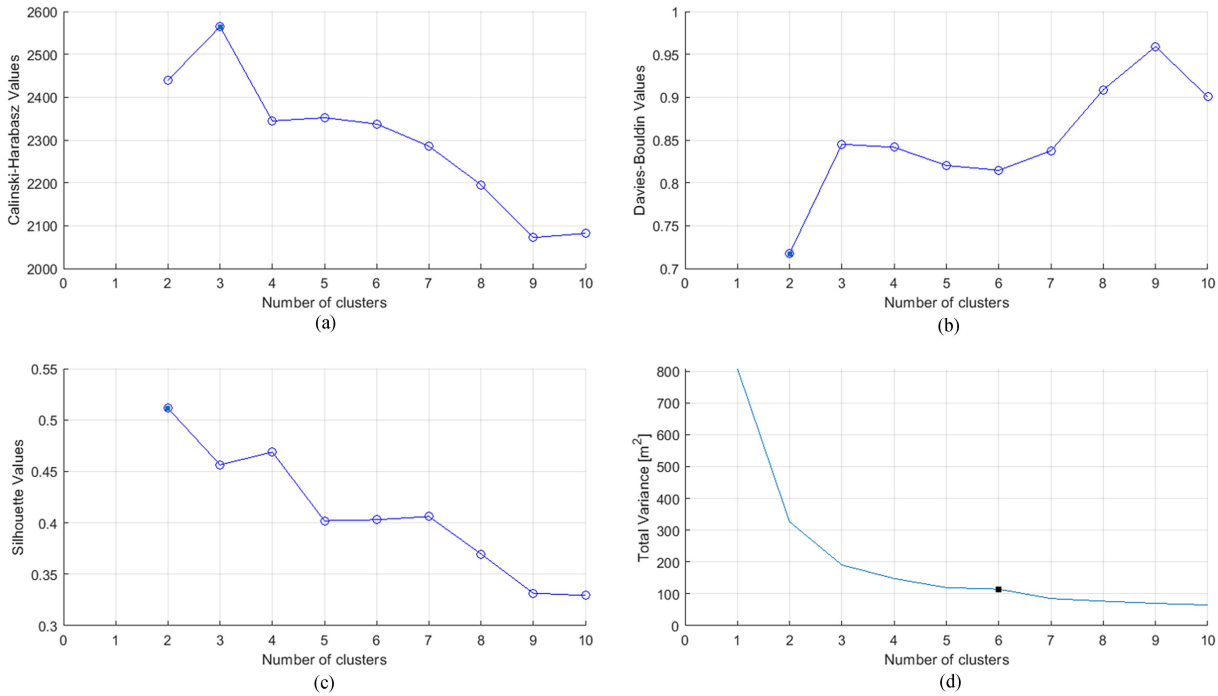


Fig. 6. Trend of the values obtained from the criteria: (a) Calisnki–Harabasz, (b) Davies–Bouldin, (c) Silhouette, and (d) Total variance as the number of clusters increases for Rada di Santa Panagia. The black point identifies the optimal number of clusters.

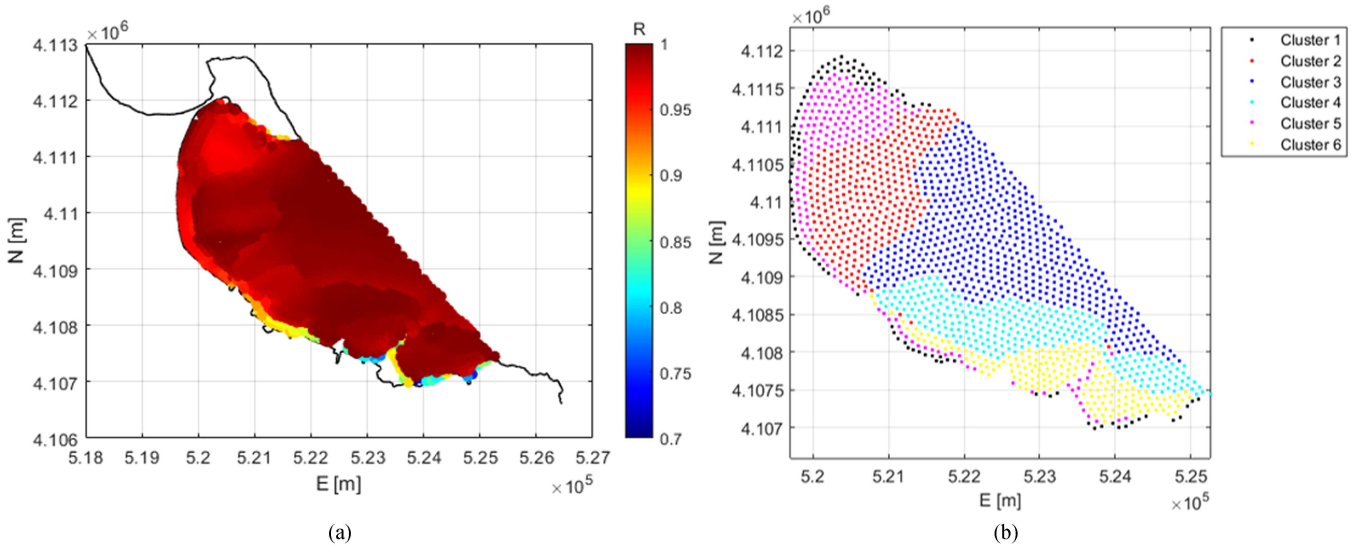


Fig. 7. (a) Distribution of the values of R and (b) application of the K-means with six clusters for Rada di Santa Panagia.

lower performance indexes are observed when considering only a few clusters, indicating nonhomogeneous areas in terms of Hs. This observation underscores that the Total Variance criterion is preferable and it stands out as the most reliable parameter in the overall performance analysis.

Table IV summarized the results obtained by the performance index R for each area. In particular, for Rada di Santa Panagia, among the different numbers of clusters tested, the higher R value of 0.99 was observed for nine clusters. However, in order to select the optimal number of clusters in an impartial way, the percentage difference between the Total Variance values of two consecutive sets of clusters was calculated and the first value, which reached less than 10%, was chosen. The percentage error is defined as

$$\%\text{Error} = \frac{|\text{TV}_i - \text{TV}_{i+1}|}{\text{TV}_i} \times 100 \tag{5}$$

where TV_i is the value obtained by the Total Variance criterion when i clusters are considered.

TABLE IV
PERFORMANCE INDEX R CALCULATED FOR EACH AREA TO FIND THE OPTIMAL NUMBER OF CLUSTERS

N. clusters	Costa Pergola	Porto Xifonio	Porto Megarese	Seno di Priolo	Rada di Santa Panagia	Baia di Siracusa
2	0.995	0.973	0.896	0.979	0.965	0.987
3	0.995	0.982	0.905	0.985	0.977	0.994
4	0.995	0.983	0.900	0.982	0.980	0.996
5	0.997	0.982	0.928	0.984	0.988	0.994
6	0.998	0.985	0.935	0.983	0.981	0.996
7	0.997	0.989	0.951	0.986	0.990	0.996
8	0.998	0.986	0.952	0.989	0.989	0.996
9	0.998	0.986	0.954	0.991	0.991	0.996
10	0.998	0.988	0.964	0.991	0.990	0.996

According to that, the optimal number for Rada di Santa Panagia was set equal to 6. It is also confirmed by the insignificant variation of the values of the performance indexes when this value is reached.

The demonstration of the application of the K-means algorithm with six clusters and the corresponding spatial distribution of R for Rada di Santa Panagia are illustrated in Fig. 7(a) and (b), respectively. The results indicate that R values are close to 1 throughout the area, except near the coastline where shoaling processes have a significant impact.

V. CONCLUSION

In this article, we introduce a novel methodology for clustering Hs with the objective of optimizing the use of ANNs to forecast and nowcast nearshore wave characteristics under actual conditions. The methodology is based on the reconstruction of wave climate characteristics within the port of Augusta (Sicily) and in the implementation of the K-means algorithm in order to examine extensive areas with diverse physical conditions, by minimizing the computational times.

The first step of the proposed methodology involves examining the wind and wave characteristics in the study area in order to establish a comprehensive database of possible scenarios. Using the model SWAN, wave climate has been propagated to obtain information in the port area. The second step was related to the clustering of the area, initially with an expert approach and then with the K-means algorithm. The expert analysis emphasized the distinct morphological conditions exhibited by the port of Augusta, rendering it a compelling case study where wave propagation phenomena are clearly evident. Thus, eight homogeneous areas (Costa Pergola, Porto Xifonio, Porto Megarese, Seno di Priolo, Rada di Santa Panagia, Baia di Siracusa, Bathymetry deeper than 100, and points outside the breakwaters) were identified. The K-means algorithm was then applied to each macro-area. Various evaluations about the distance metrics and the optimal number of clusters were made to find the best configuration of the algorithm. The comparison of four distance metrics for Rada di Santa Panagia showed that City-block and Euclidean distances better represent the behaviour of Hs. Moreover, the performance indexes calculated for each metric confirmed the best results provided by these two distances. The second comparison was made between four criteria to find the optimal number of clusters. Specifically, for the Rada di Santa Panagia, utilising the Total Variance criterion yields more significant results by indicating that increasing the number of clusters beyond six is unlikely to improve the clustering performance substantially. This finding aligns with the values obtained from the calculated performance indexes, confirming the validity of the Total Variance criterion.

In conclusion, the proposed methodology is promising, since clustering is able to effectively capture the Hs behavior. However, the analysis reveals that the choice of distance metrics can significantly impact the performance of standard clustering algorithms such as K-means. Future work will be dedicated to refining and optimizing the proposed methodology, with a specific focus on neural network applications to assess nearshore wave characteristics under real-world conditions within each identified cluster. This entails developing innovative strategies to enhance the performance and accuracy of neural network models in predicting nearshore wave patterns across diverse morphological settings. The ultimate objective is to construct a comprehensive forecasting map that encompasses the entire study area, incorporating detailed information on nearshore wave behavior and associated risk factors. This forecasting map will serve as a valuable tool for maritime stakeholders, providing them with actionable insights to improve decision-making and mitigate potential hazards.

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